

39번

When we practice, we want to strengthen the synapses relaying information on how to play something correctly, while weakening those that send erroneous messages.

To understand how this works, think of a leaky hose that has many holes in it.

Some of the water will go through the hose and out the nozzle, but a lot of it will leak out the holes.

This is what your brain is like when you first start to learn something: the water running out the holes is all the erroneous information your brain is sending to your fingers, lips, etc.

Once you plug the holes in the hose, all the water goes then out the nozzle; in your brain, this is analogous to the synapses relaying the correct messages being much stronger than those sending incorrect messages.

The brain accomplishes this through changing the structure of the synapse to make it easier for the correct neurons to communicate.

A common phrase in neuroscience is "neurons that fire together wire together," meaning neurons that communicate with each other often change their structure to make that communication easier.